
MONGO ARCHIVER v6 – COMPLETE RELEASE

June 2, 2025

WHAT YOU GET

1. mongo-archiver-v6-release.zip (37 KB)
 - ├─ codebase/ ← Complete Spring Boot project (ready to build)
 - ├─ README.md ← Quick start guide
 - ├─ MIGRATION_PLAN.txt ← Step-by-step deployment & monitoring
 - ├─ DOCUMENTATION.txt ← Complete technical documentation
 - └─ flow_diagram.dot ← Graphviz diagram source
2. flow_diagram.jpg (103 KB)

Visual flowchart of the entire archival process.

Shows: startup → backfill detection → monthly slices → daily scheduler
3. MIGRATION_PLAN.txt (9.8 KB)

Standalone guide for deployment, monitoring, and troubleshooting.
4. mongo-archiver-v6-complete.txt (56 KB)

Full technical documentation with complete source code.

QUICK START (3 STEPS)

1. Unzip the release:

```
unzip mongo-archiver-v6-release.zip
cd mongo-archiver-release/codebase
```
2. Build:

```
mvn clean package -DskipTests
```
3. Run (first time):

```
MONGO_URI=mongodb://your-host:27017 \
MONGO_DB=yourdb \
java -jar target/mongo-archiver-1.0.0.jar
```

The app will:

- Automatically detect if backfill is needed
- Run monthly slices (e.g., 36 months of data)
- Save progress after each slice
- Resume on restart (no manual intervention)
- Hand over to daily scheduler when done

Done. The archiver is now running.

KEY FEATURES

✓ AUTOMATED BACKFILL

No manual monthly slices. App detects oldest record, calculates slices, runs them automatically.

✓ RESTART-SAFE

If app crashes mid-backfill, next startup resumes from the last completed slice. No loss of progress.

✓ ZERO MANUAL STEPS

One command to deploy. Everything else is automatic.

✓ PER-ENTITY AUDIT LOG

One document per entity per run with all collection outcomes.

✓ ANCHOR-LAST STRATEGY

If a related collection fails, anchor stays in source for automatic retry.

✓ CONFIGURABLE

Just edit application.yml. No Java code changes needed to add new entities.

WHAT'S INSIDE THE ZIP

mongo-archiver-release/

├─ README.md (133 lines)

| Quick start, file structure, key features, configuration guide.

|

├─ MIGRATION_PLAN.txt (285 lines)

| Step-by-step deployment, monitoring queries, troubleshooting.

| → Start here for deployment guidance.

```

|
├─ DOCUMENTATION.txt (1430 lines)
|   Complete technical reference. Architecture, code walkthrough,
|   flow diagrams, audit log schema, monitoring queries.
|   → Read this for deep understanding.
|
├─ flow_diagram.dot
|   Graphviz source. Generate JPG with:
|   dot -Tjpg flow_diagram.dot -o diagram.jpg
|
└─ codebase/ (Maven project)
    ├─ pom.xml (Spring Boot 3.2.3, Java 17, MongoDB)
    |
    ├─ src/main/resources/
    |   └─ application.yml
    |       Configure schedule, timezone, jobs, retention window.
    |
    └─ src/main/java/com/archiver/
        ├─ MongoArchiverApplication.java (entry point)
        ├─ config/ArchiverProperties.java
        ├─ core/
        |   ├─ Archiver.java (archive engine)
        |   ├─ BackfillService.java (orchestrates startup backfill)
        |   └─ BackfillStateRepository.java (persists progress)
        ├─ job/ArchiveCronJob.java (scheduler)
        ├─ model/
        |   ├─ ArchiveLogEntry.java (audit log)
        |   └─ BackfillState.java (backfill progress)
        └─ util/
            ├─ ArchiveLogger.java
            ├─ IdPathResolver.java
            └─ IndexManager.java

```

CONFIGURATION

Edit codebase/src/main/resources/application.yml:

```

spring.data.mongodb.uri: your MongoDB connection
spring.data.mongodb.database: your database name

```

```

archiver.schedule: "0 0 2 * * ?"      (2 AM UTC)
archiver.timezone: Asia/Kolkata      (for IST)

```

```

archiver.jobs:

```

```

- name: orders
  batch-size: 100
  anchor:
    collection: orders
    id-field: orderId
    created-ts-field: createdAt
    archive-after-days: 180          (6 months)
    archive-collection: orders_archive
  related:
    - collection: orderItems
      id-path: orderId              (supports dot-notation for nested fields)
      archive-collection: orderItems_archive

```

Add more jobs the same way. No Java code changes needed.

MONITORING

During backfill:

```

db.backfillState.find({}, { jobName:1, status:1, completedSlices:1,
totalSlices:1 })

```

Check for failures:

```

db.archiveLog.find({ status: { $in: ["FAILED","PARTIAL_FAILURE"] } })

```

Full details:

→ See MIGRATION_PLAN.txt section "MONITORING QUERIES"

WHAT CHANGED FROM v5

v5 required manual env vars (ARCHIVE_FROM_DATE / ARCHIVE_TO_DATE) to slice the backfill. This was still manual.

v6 AUTOMATES THE ENTIRE BACKFILL:

- BackfillService detects on startup if backfill is needed
- Calculates monthly slices automatically
- Runs them sequentially with progress saved after each slice
- Restarts resume from the last completed slice
- No env vars, no manual commands needed

Result: One deployment, everything happens automatically.

BUILD & DEPLOY

Prerequisites:

- Java 17+
- Maven 3.8+
- MongoDB 4.2+ with replica set (transactions needed)

Build:

```
mvn clean package -DskipTests
```

Run:

```
java -jar target/mongo-archiver-1.0.0.jar
```

For immediate (non-scheduled) run:

```
java -jar target/mongo-archiver-1.0.0.jar --run-now
```

DIAGRAM

flow_diagram.jpg shows the complete flow:

APP STARTUP

- BackfillService checks if backfill is needed
- If yes: calculates and runs monthly slices
- If no: skips to scheduler
- Scheduler becomes active for daily runs

TROUBLESHOOTING

"Backfill is running too slow"

- Reduce batch-size in application.yml (e.g., 100 → 50)
- Restart (resumes from last completed slice)

"A backfill slice failed"

- Check logs: `tail -f logs/archiver.log`
- Check archiveLog: `db.archiveLog.find({ status: "FAILED" })`
- Fix the underlying issue, restart
- App will resume from the failed slice

"I want to restart backfill from scratch"

→ db.backfillState.deleteOne({ jobName: "orders" })
→ Restart the app

For more:

→ See MIGRATION_PLAN.txt section "WHAT IF SOMETHING FAILS"

GETTING HELP

For quick deployment steps:

→ README.md in the zip

For detailed migration & monitoring:

→ MIGRATION_PLAN.txt

For architecture & code details:

→ DOCUMENTATION.txt

For visual overview:

→ flow_diagram.jpg

VERSION HISTORY

v1 - Basic archiver with timestamp cutoff
v2 - Added temporary index management
v3 - One audit log entry per entity (instead of per entity×collection)
v4 - Added per-collection result map to audit log
v5 - Added date-range backfill via env vars (manual slicing)
v6 - AUTOMATED BACKFILL ON STARTUP with resume-on-restart ← YOU ARE HERE

END OF RELEASE SUMMARY
